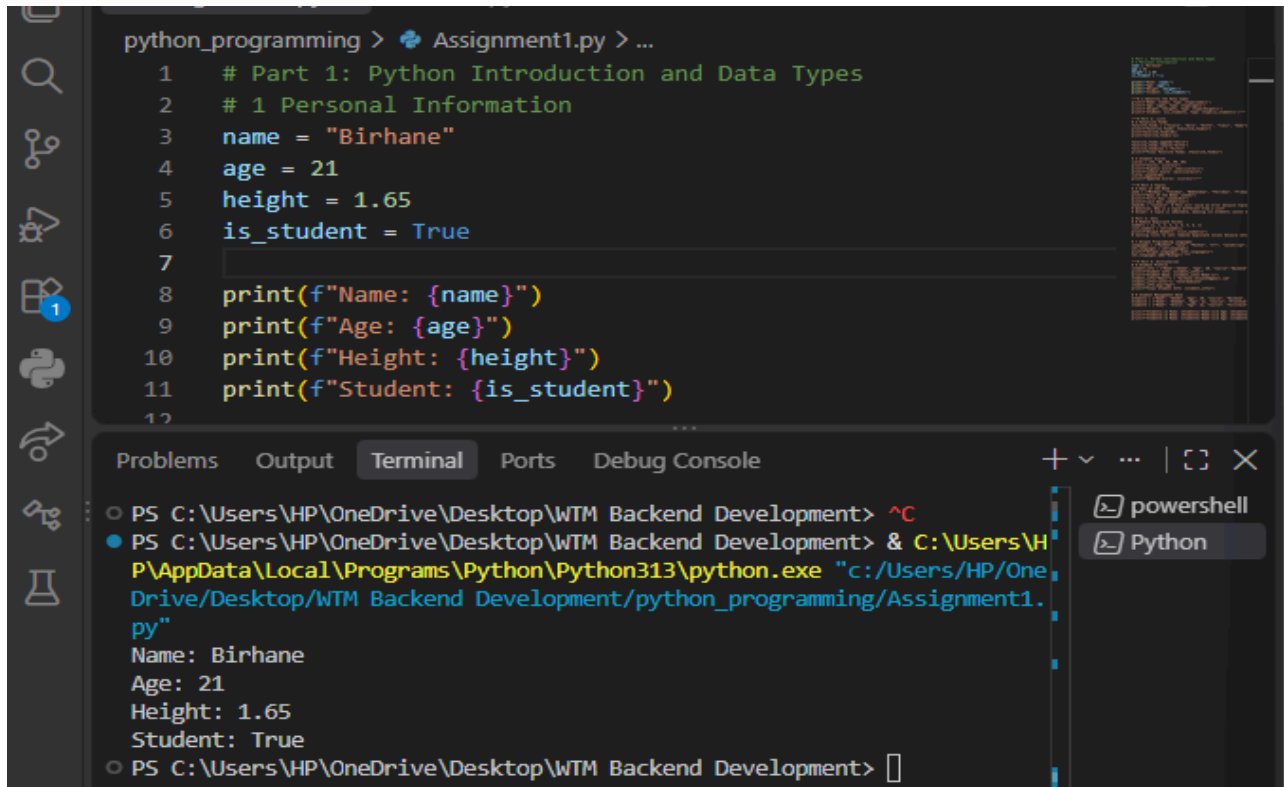


Part 1: Python Introduction and Data Type

1. Personal Information



```
python_programming > Assignment1.py > ...
1  # Part 1: Python Introduction and Data Types
2  # 1 Personal Information
3  name = "Birhane"
4  age = 21
5  height = 1.65
6  is_student = True
7
8  print(f"Name: {name}")
9  print(f"Age: {age}")
10 print(f"Height: {height}")
11 print(f"Student: {is_student}")
12
```

Problems Output Terminal Ports Debug Console

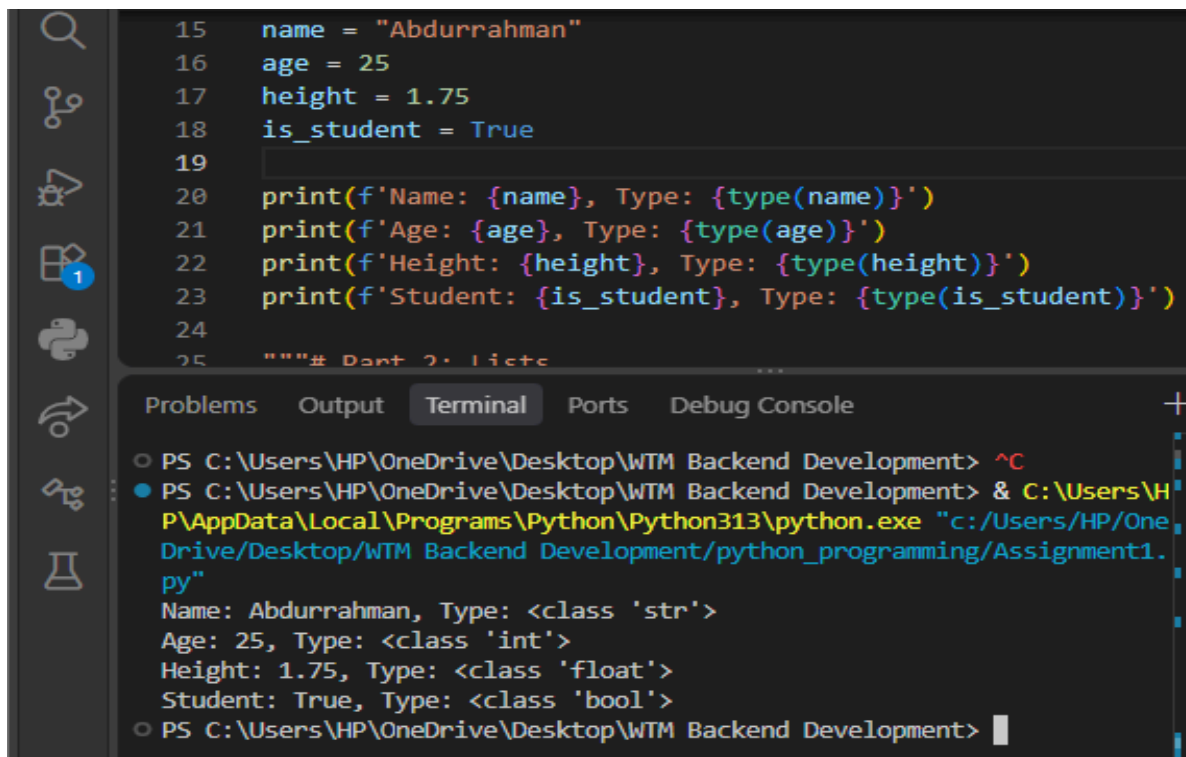
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C

PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Assignment1.py"

Name: Birhane
Age: 21
Height: 1.65
Student: True

PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>

2. Identify the Data Types



```
15 name = "Abdurrahman"
16 age = 25
17 height = 1.75
18 is_student = True
19
20 print(f'Name: {name}, Type: {type(name)}')
21 print(f'Age: {age}, Type: {type(age)}')
22 print(f'Height: {height}, Type: {type(height)}')
23 print(f'Student: {is_student}, Type: {type(is_student)}')
24
25 """# Part 2: Lists
```

Problems Output Terminal Ports Debug Console

PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C

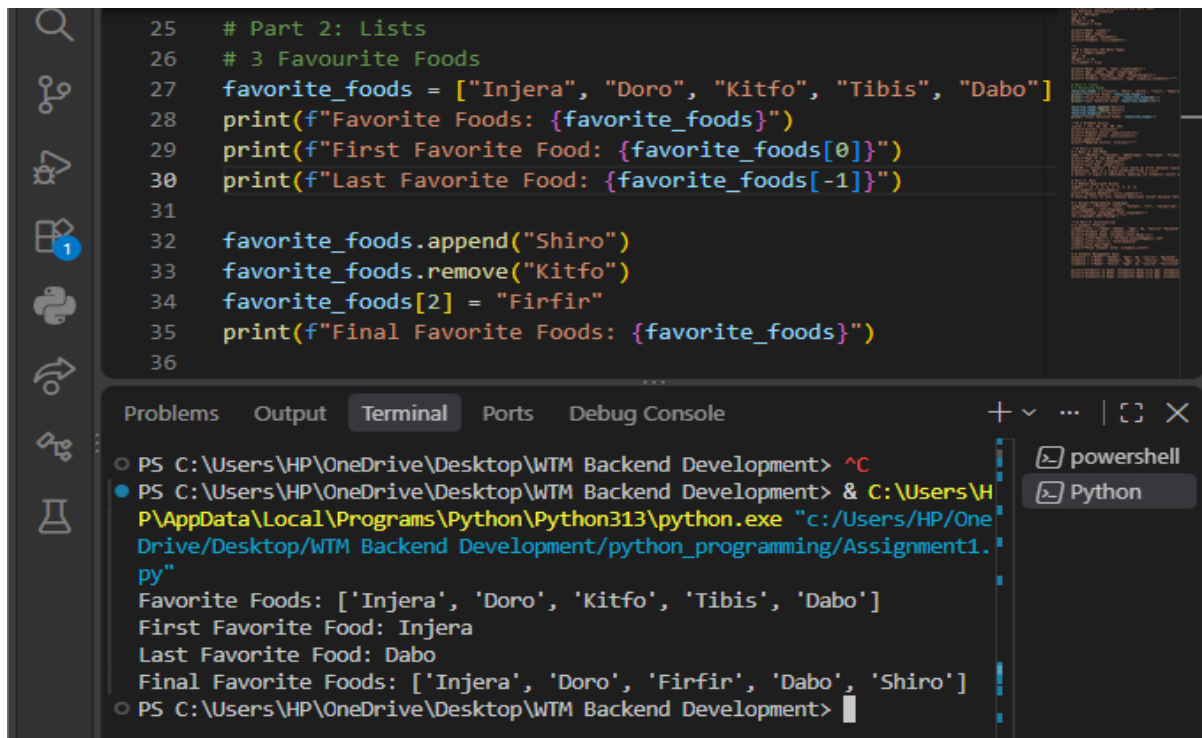
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Assignment1.py"

Name: Abdurrahman, Type: <class 'str'>
Age: 25, Type: <class 'int'>
Height: 1.75, Type: <class 'float'>
Student: True, Type: <class 'bool'>

PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>

Part 2: Lists

3. Favourite Foods

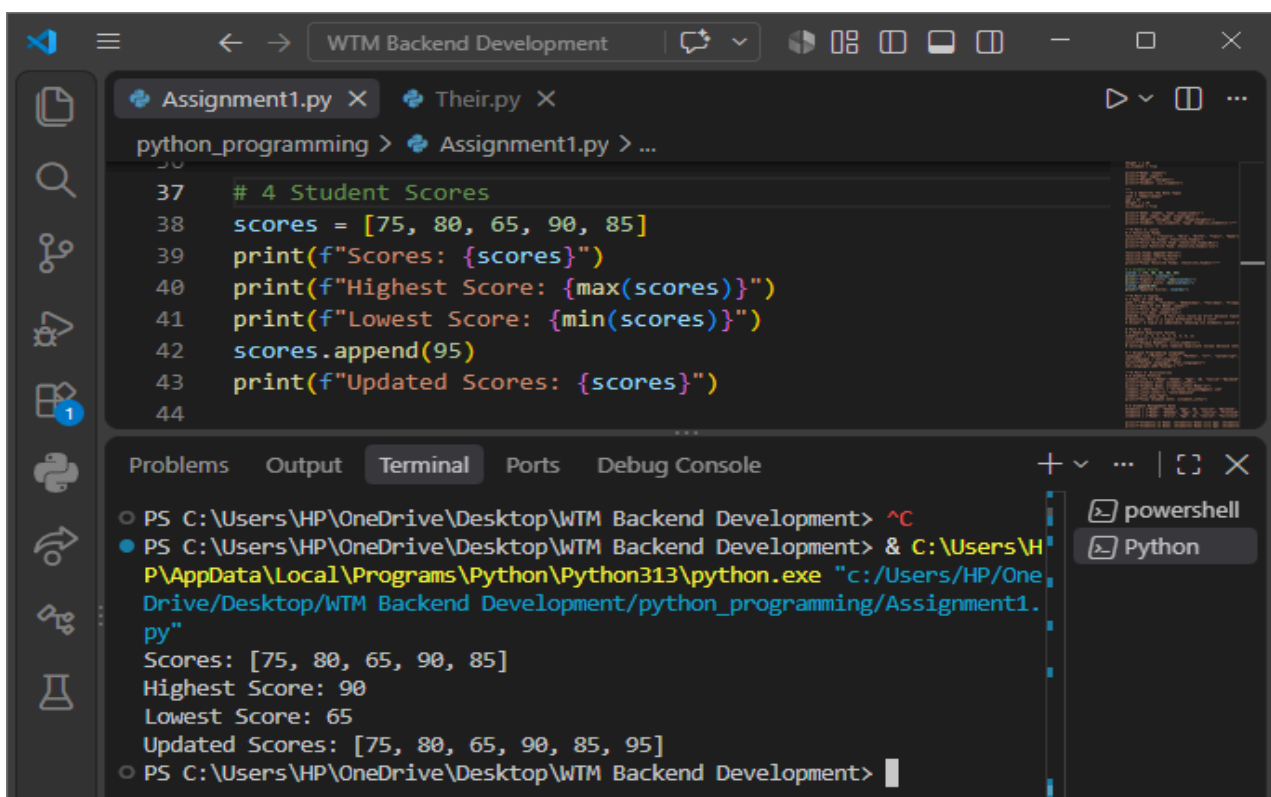


```
25 # Part 2: Lists
26 # 3 Favourite Foods
27 favorite_foods = ["Injera", "Doro", "Kitfo", "Tibis", "Dabo"]
28 print(f"Favorite Foods: {favorite_foods}")
29 print(f"First Favorite Food: {favorite_foods[0]}")
30 print(f"Last Favorite Food: {favorite_foods[-1]}")
31
32 favorite_foods.append("Shiro")
33 favorite_foods.remove("Kitfo")
34 favorite_foods[2] = "Firfir"
35 print(f"Final Favorite Foods: {favorite_foods}")
36
```

Terminal Output:

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\OneDrive\Desktop\WTM Backend Development\python_programming\Assignment1.py
Favorite Foods: ['Injera', 'Doro', 'Kitfo', 'Tibis', 'Dabo']
First Favorite Food: Injera
Last Favorite Food: Dabo
Final Favorite Foods: ['Injera', 'Doro', 'Firfir', 'Dabo', 'Shiro']
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>
```

4. Student Scores



```
37 # 4 Student Scores
38 scores = [75, 80, 65, 90, 85]
39 print(f"Scores: {scores}")
40 print(f"Highest Score: {max(scores)}")
41 print(f"Lowest Score: {min(scores)}")
42 scores.append(95)
43 print(f"Updated Scores: {scores}")
44
```

Terminal Output:

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\OneDrive\Desktop\WTM Backend Development\python_programming\Assignment1.py
Scores: [75, 80, 65, 90, 85]
Highest Score: 90
Lowest Score: 65
Updated Scores: [75, 80, 65, 90, 85, 95]
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>
```

Part 3: Tuples

5. Days of the Week

```
45 # Part 3 Tuples
46 # 5 Days of the Week
47 week = ("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday")
48 print(f"Days of the Week: {week}")
49 print(f"First Day: {week[0]}")
50 print(f>Last Day: {week[-1]}")
51 week[0] = "January" # This will raise an error because tuples are immutable
52 # Question: Why is a tuple different from a list?
53 # Answer: A tuple is immutable, meaning its elements cannot be changed
54 # after creation, while a list is mutable and allows modifications.
55
```

Problems Output Terminal Ports Debug Console

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Assignment1.py"
Days of the Week: ('Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday')
First Day: Monday
Last Day: Sunday
Traceback (most recent call last):
  File "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Assignment1.py", line 51, in <module>
    week[0] = "January" # This will raise an error because tuples are immutable
    ~~~~~^^^
TypeError: 'tuple' object does not support item assignment
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>
```

Part 4: Sets

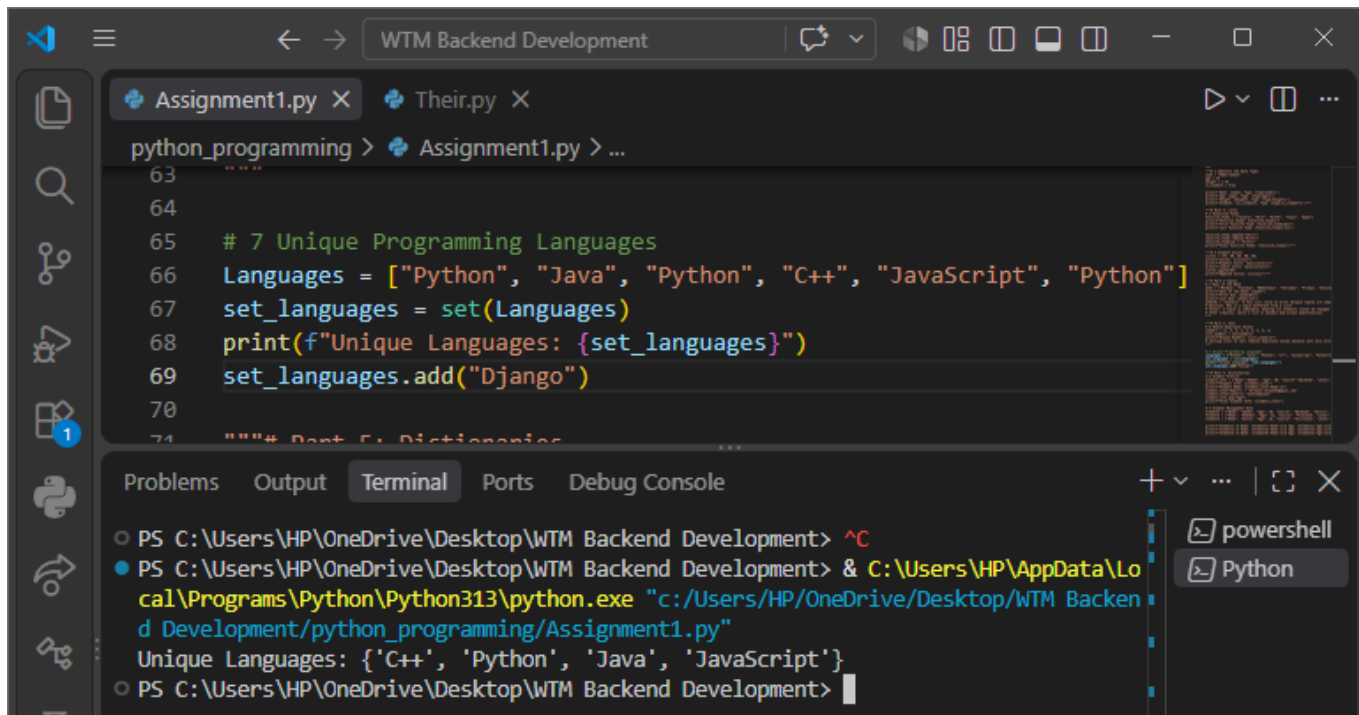
6. Remove Duplicate Values

```
56
57 # Part 4: Sets
58 # 6 Remove Duplicate Values
59 numbers = [1, 2, 3, 4, 2, 5, 3, 6, 1]
60 list_numbers = set(numbers)
61 print(f"Unique Numbers: {list_numbers}")
62 # Casting lists to sets removes duplicate values because sets only allow unique values
63
```

Problems Output Terminal Ports Debug Console

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Assignment1.py"
Unique Numbers: {1, 2, 3, 4, 5, 6}
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>
```

7. Unique Programming Languages



The screenshot shows a Visual Studio Code editor window titled "WTM Backend Development". The editor has two tabs open: "Assignment1.py" and "Their.py". The "Assignment1.py" tab is active, showing a Python script. The script defines a list of programming languages, converts it to a set, prints the unique languages, and adds "Django" to the set. The terminal at the bottom shows the command to run the script, and the output displays the unique languages: C++, Python, Java, and JavaScript.

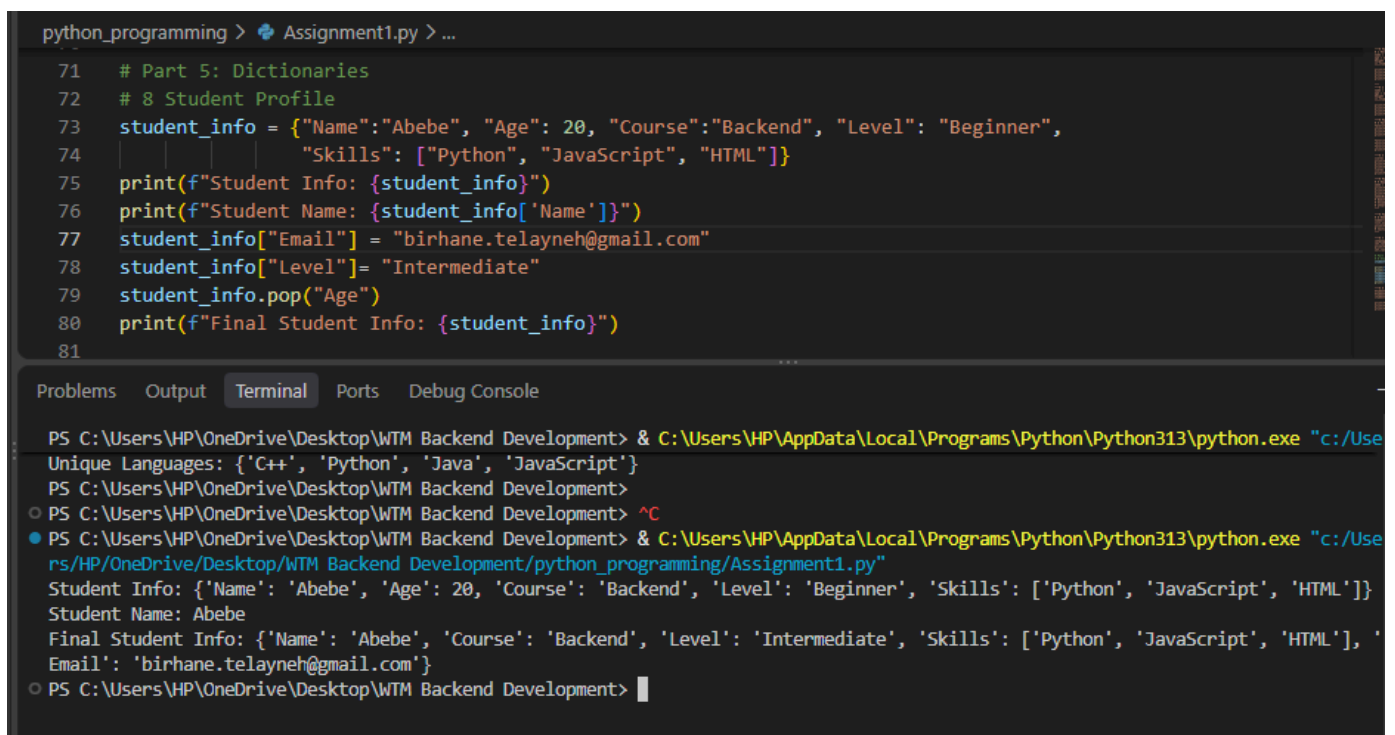
```
python_programming > Assignment1.py > ...
63
64
65 # 7 Unique Programming Languages
66 Languages = ["Python", "Java", "Python", "C++", "JavaScript", "Python"]
67 set_languages = set(Languages)
68 print(f"Unique Languages: {set_languages}")
69 set_languages.add("Django")
70
71 """# Part 5: Dictionaries
```

Terminal Output:

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Assignment1.py"
Unique Languages: {'C++', 'Python', 'Java', 'JavaScript'}
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>
```

Part 5: Dictionaries

8. Student Profile



The screenshot shows a Visual Studio Code editor window titled "WTM Backend Development". The editor has one tab open: "Assignment1.py". The script defines a dictionary for a student's profile, prints the dictionary, the student's name, updates the email and level, removes the age, and prints the final dictionary. The terminal at the bottom shows the command to run the script, and the output displays the student's profile information.

```
python_programming > Assignment1.py > ...
71 # Part 5: Dictionaries
72 # 8 Student Profile
73 student_info = {"Name": "Abebe", "Age": 20, "Course": "Backend", "Level": "Beginner",
74               "Skills": ["Python", "JavaScript", "HTML"]}
75 print(f"Student Info: {student_info}")
76 print(f"Student Name: {student_info['Name']}")
77 student_info["Email"] = "birhane.telayneh@gmail.com"
78 student_info["Level"] = "Intermediate"
79 student_info.pop("Age")
80 print(f"Final Student Info: {student_info}")
81
```

Terminal Output:

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Assignment1.py"
Student Info: {'Name': 'Abebe', 'Age': 20, 'Course': 'Backend', 'Level': 'Beginner', 'Skills': ['Python', 'JavaScript', 'HTML']}
Student Name: Abebe
Final Student Info: {'Name': 'Abebe', 'Course': 'Backend', 'Level': 'Intermediate', 'Skills': ['Python', 'JavaScript', 'HTML'], 'Email': 'birhane.telayneh@gmail.com'}
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>
```

9. Student Management Data

```
81
82 # 9 Student Management Data
83 Student1 = {"Name": "Abebe", "Age": 20, "Course": "Backend", "Skills": ["Python", "JavaScript", "HTML"]}
84 Student2 = {"Name": "Kebede", "Age": 21, "Course": "Frontend", "Level": "Intermediate", "Skills": ["CSS", "JavaScript", "React"]}
85 student3 = {"Name": "Glory", "Age": 22, "Course": "Fullstack", "Level": "Advanced", "Skills": ["Python", "Django", "React"]}
86
87 print(f"Student1:\n Name: {Student1['Name']}\n Age: {Student1['Age']}\n Course: {Student1['Course']}\n Skills: {Student1['Skills'][0]}, {Student1['Skills'][1]}, {Student1['Skills'][2]}")
88 print(f"Student2:\n Name: {Student2['Name']}\n Age: {Student2['Age']}\n Course: {Student2['Course']}\n Skills: {Student2['Skills'][0]}, {Student2['Skills'][1]}, {Student2['Skills'][2]}")
89 print(f"Student3:\n Name: {student3['Name']}\n Age: {student3['Age']}\n Course: {student3['Course']}\n Skills: {student3['Skills'][0]}, {student3['Skills'][1]}, {student3['Skills'][2]}")
90
```

Problems Output Terminal Ports Debug Console

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Assignment1.py"
Student1:
Name: Abebe
Age: 20
Course: Backend
Skills: Python, JavaScript, HTML
Student2:
Name: Kebede
Age: 21
Course: Frontend
Skills: CSS, JavaScript, React
Student3:
Name: Glory
Age: 22
Course: Fullstack
Skills: Python, Django, React
```

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